# ECE 3355: Electronics

Section 16721
Fall 2018

## Exam 2

Version B - Fall 2018

**November 15, 2018**

Do *not* open the exam until instructed to do so. Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page. This is a closed-book/notes exam and you *may* use a calculator. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: _____

| Question | Points | Score |
|---|---|---|
| 1 | 25 | |
| 2 | 30 | |
| 3 | 45 | |
| Total: | 100 | |

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace $v_{BE}$ with $v_{EB}$.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T =$ thermal voltage $= \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 6.4 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

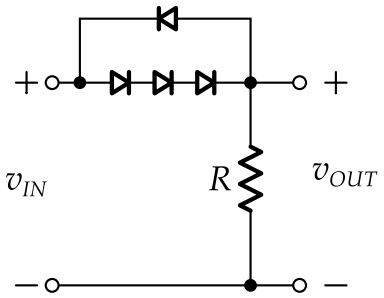
In Terms of $g_m$

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

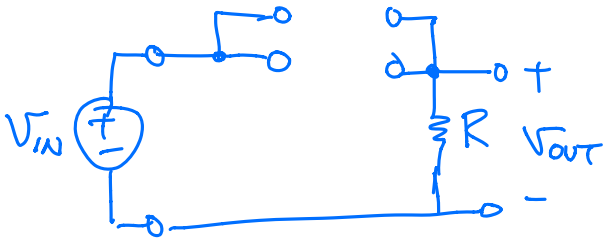
In Terms of $r_e$

$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

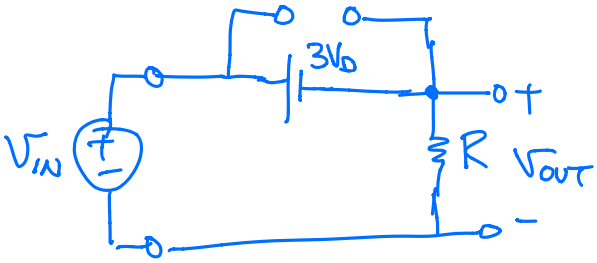
1. For the circuit below, assume $V_D = 0.7$ volts and $r_D = 0$.



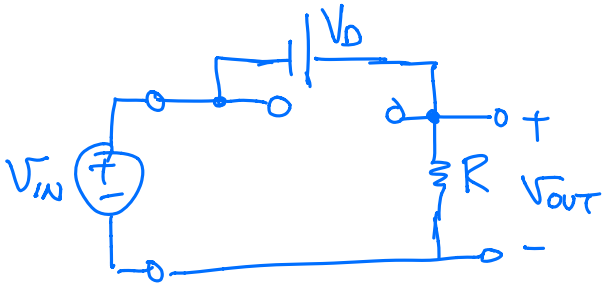

(a) (10 points) Write an expression for $v_{OUT}$ in terms of $v_{IN}$.




when $-V_D < V_{IN} < 3V_D$, both sets of diodes are off and $V_{OUT} = 0$



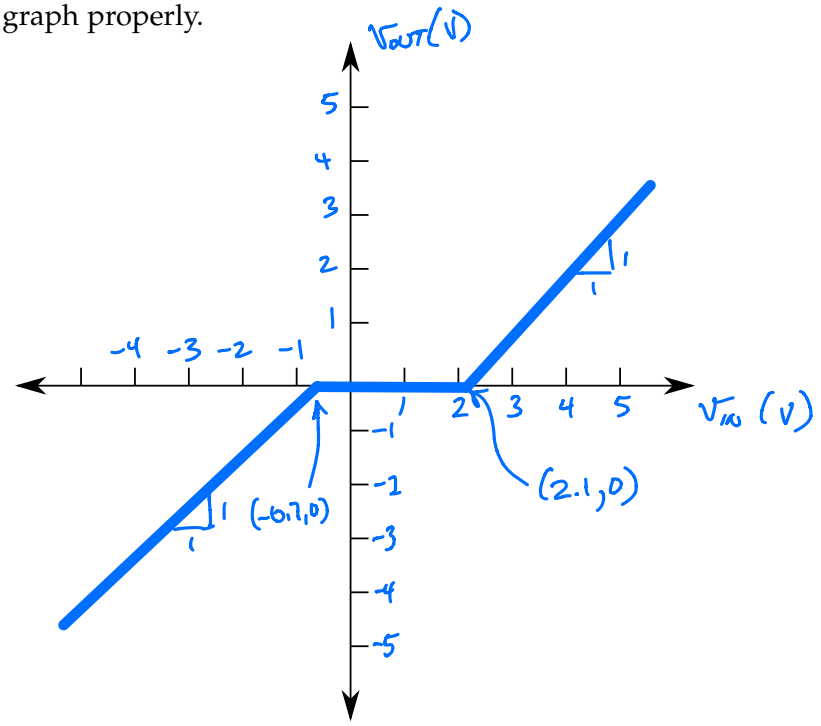

when $V_{IN} > 3V_D$, the three diodes turn on and $V_{OUT} = V_{IN} - 3V_D$



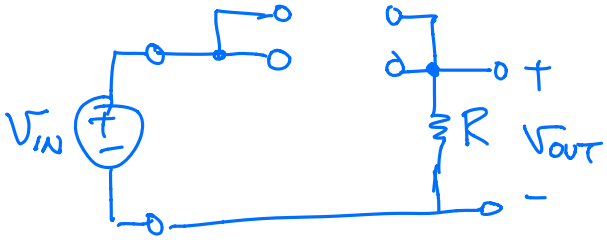

when $V_{IN} < -V_D$, the single diode is forward-biased, and $V_{OUT} = V_{IN} + V_D$

$$V_{OUT} = \begin{cases} V_{IN} + V_D & V_{IN} < -V_D \\ 0 & -V_D \le V_{IN} < 3V_D \\ V_{IN} - 3V_D & V_{IN} \ge 3V_D \end{cases}$$

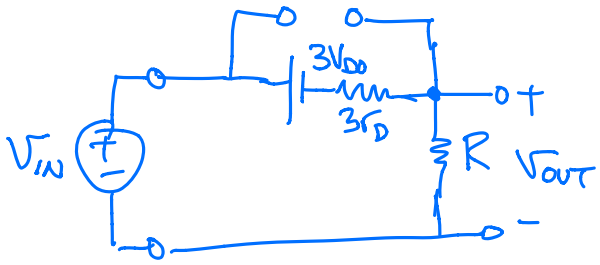
(b) (5 points) Sketch $v_{OUT}$ and $v_{IN}$ for $-4$ volts $< v_{IN} < 4$ volts on the graph provided. Be sure to label the graph properly.



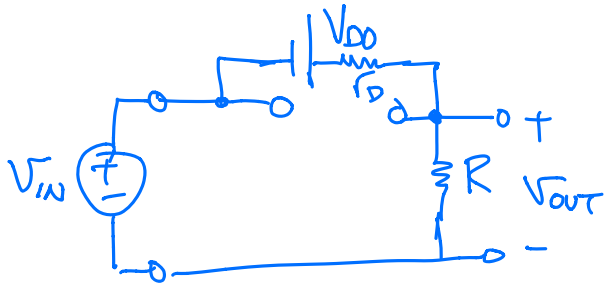

(c) (10 points) Now, assume that $r_D$ is no longer zero, $R = r_D$, and $V_{D0} = 0.5$ volts. Write an expression for $v_{OUT}$ in terms of $v_{IN}$.




$$V_{OUT} = \begin{cases} \frac{V_{IN} + V_{D0}}{2} & V_{IN} < -V_{D0} \\ 0 & -V_{D0} < V_{IN} < 3V_{D0} \\ \frac{V_{IN} - 3V_{D0}}{4} & V_{IN} > 3V_{D0} \end{cases}$$

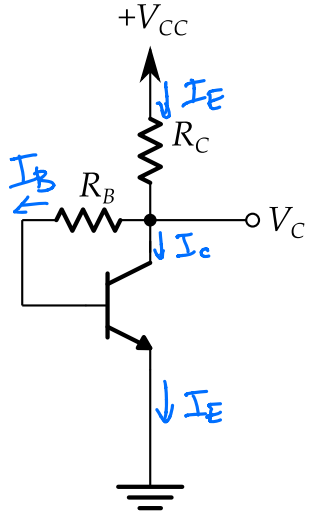



$$V_{OUT} = \frac{V_{IN} - 3V_{D0}}{R + 3r_D} \cdot R = \frac{V_{IN} - 3V_{D0}}{4}$$




$$V_{OUT} = \frac{V_{IN} + V_{D0}}{r_D + R} \cdot R = \frac{V_{IN} + V_{D0}}{2}$$

2. Use the following circuit to complete this problem.




(a) (10 points) Write expressions for $V_C$ and the currents in the transistor in terms of $R_B$, $R_C$, $V_{CC}$, $V_{BE}$, and $\beta$.

$V_{cc} = I_E R_C + I_B R_B + V_{BE}$

$I_E = (\beta+1) I_B$

$$I_B = \frac{V_{cc} - V_{BE}}{(\beta+1) R_C + R_B}$$

$I_C = \beta I_B$

$I_E = (\beta+1) I_B$

$V_C = V_{CC} - I_E R_C$

or

$V_C = V_{BE} + I_B R_B$

(b) (10 points) Find values for $R_B$ and $R_C$ so that $V_C = V_{CC}/2$ and $r_\pi = 1\,\text{k}\Omega$ when $V_{CC} = 25\,\text{V}$ and $\beta = 99$.

$$r_\pi = \frac{V_T}{I_B} \Rightarrow I_B = \frac{V_T}{r_\pi} = \frac{0.025\text{V}}{1\text{k}\Omega} = 0.025\,\text{mA}$$

$$V_C = V_{BE} + I_B R_B \Rightarrow R_B = \frac{V_C - V_{BE}}{I_B} = \frac{12.5\text{V} - 0.7\text{V}}{0.025\text{mA}} = 472\,\text{k}\Omega$$

$$V_C = V_{CC} - I_E R_C \Rightarrow R_C = \frac{V_{CC} - V_C}{(\beta+1) I_B} = \frac{25\text{V} - 12.5\text{V}}{(99+1)\,0.025\text{mA}} = 5\,\text{k}\Omega$$

(c) (10 points) Using the values for the resistors from the previous part, determine the range of $V_C$ when $50 \leq \beta \leq 150$?

$$V_C = V_{BE} + I_B R_B = V_{BE} + \frac{V_{CC} - V_{BE}}{(\beta+1)R_C + R_B} \cdot R_B$$

for $\beta = 50$

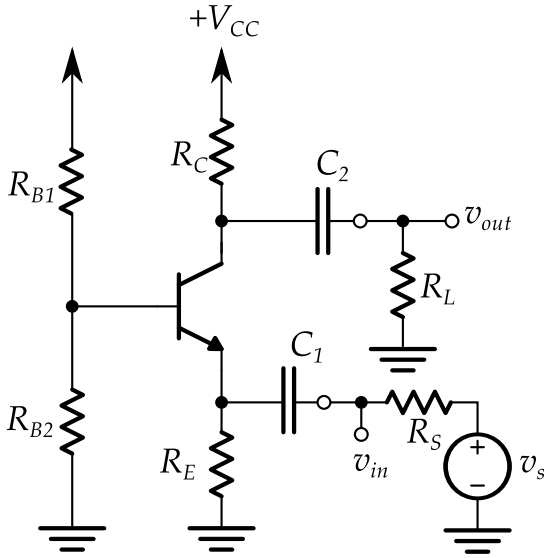
$$V_c = 0.7V + \frac{(25V - 0.7V)\cdot 472k\Omega}{(51)(5k\Omega) + 472k\Omega} = 16.5V$$

for $\beta = 151$

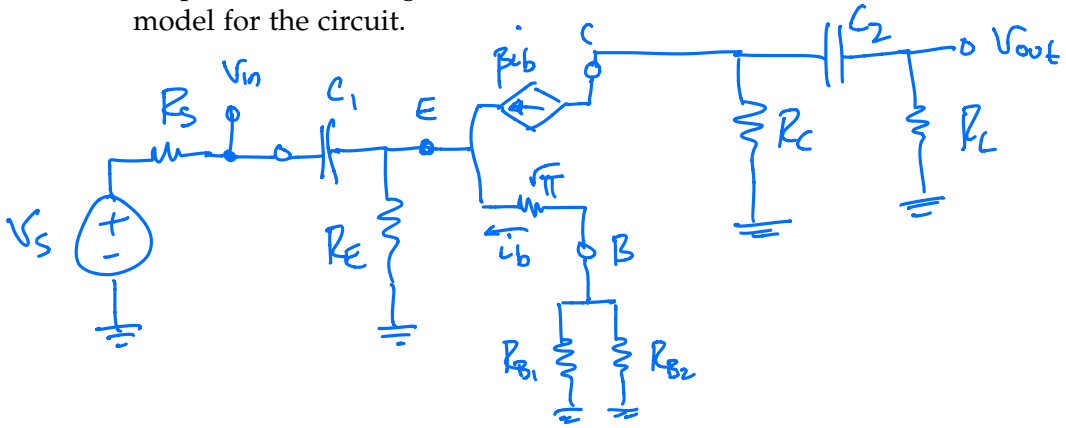
$$V_c = 0.7V + \frac{(25V - 0.7V)\cdot 472k\Omega}{(151)(5k\Omega) + 472k\Omega} = 10V$$

$$10V < V_c < 16.5V$$

3. Use the following circuit to complete this problem.

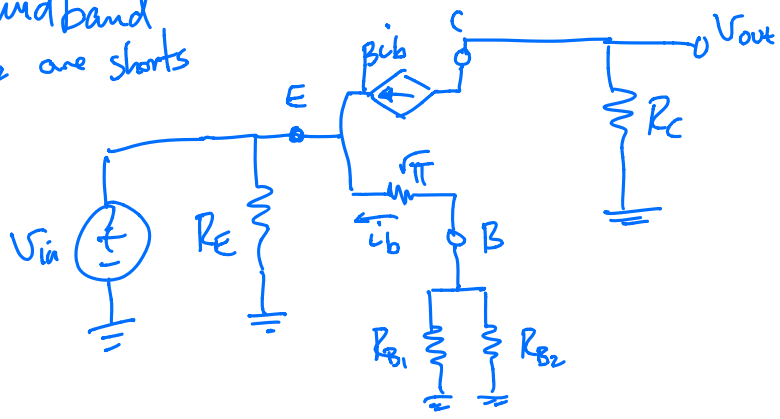



(a) (10 points) Assuming that the transistor is in the active mode, sketch the small signal model for the circuit.




(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, $\beta$, $r_\pi$.

in the midband $C_1$ & $C_2$ are shorts




$$V_{out} = -\beta i_b R_C$$

$$V_{in} = -i_b r_\pi - i_b (R_{B1} \| R_{B2})$$

$$\frac{V_{out}}{V_{in}} = \frac{\beta R_C}{r_\pi + R_{B1} \| R_{B2}}$$

(c) (10 points) Find expression for the input and output resistance for the *midband* with the source and load removed.

Two ways of solving for $R_{in}$:

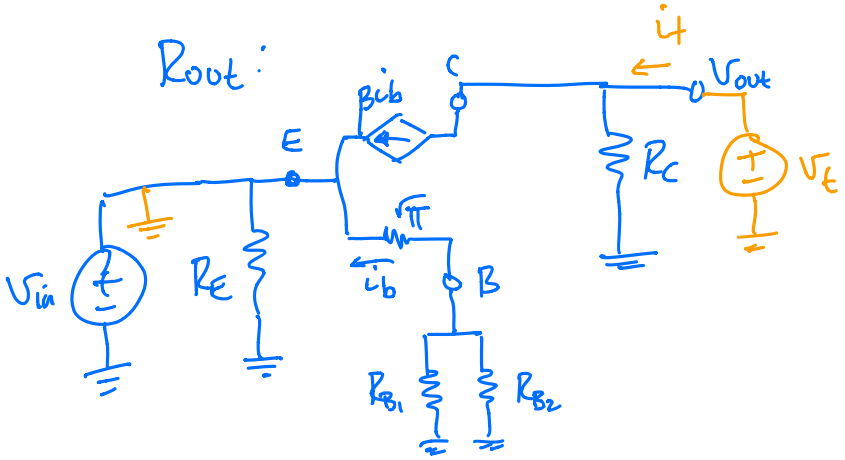
① $R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{\frac{v_{in}}{R_E} - (\beta+1) i_b} = \frac{v_{in}}{\frac{v_{in}}{R_E} - (\beta+1)\cdot\left(\frac{-v_{in}}{r_\pi + R_{B1}\|R_{B2}}\right)}$

$= \frac{1}{\frac{1}{R_E} + \frac{1}{\frac{r_\pi + R_{B1}\|R_{B2}}{\beta+1}}} = R_E \| \left[\frac{r_\pi + R_{B1}\|R_{B2}}{\beta+1}\right]$

② $R_{in} = R_E \| \frac{v_{in}}{(\beta+1) i_b} = R_E \| \frac{v_{in}}{(\beta+1)\frac{-v_{in}}{(r_\pi + R_{B1}\|R_{B2})}}$

$= R_E \| \left[\frac{r_\pi + R_{B1}\|R_{B2}}{\beta+1}\right]$

$R_{out}$:




$R_{out} = \frac{v_t}{i_t}$

Since $i_b = 0$

$R_{out} = R_C$

(d) (5 points) Write expressions for the resistances seen by the capacitors $C_1$ and $C_2$.

$C_1$ will see $R_S + R_{in}$

$C_2$ will see $R_{out} + R_L$

(e) (10 points) If a capacitor is placed in parallel with $R_{B2}$, would you treat it as a short or an open when computing the *midband* gain? Write the expression for the gain in the midband with this new capacitor.

A capacitor across $R_{B2}$ will short $R_{B1} \| R_{B2}$ at high frequencies. Since

$$\frac{v_{out}}{v_{in}} = \frac{\beta R_C}{r_\pi + R_{B1} \| R_{B2}}$$

when the capacitor is an open, it will become

$$\frac{v_{out}}{v_{in}} = \frac{\beta R_C}{r_\pi}$$

when it is a short. Hence, the capacitor is a short in the mid band since the gain is higher and

$$\frac{v_{out}}{v_{in}} = \frac{\beta R_C}{r_\pi}$$